

ASSIGNMENT-1

1. What is Artificial Intelligence (AI)?

AI is the simulation of human intelligence in machines that are programmed to think, learn, and solve problems.

2. What is the core idea of AI?

The core idea is to make machines “smart” enough to perform tasks that humans traditionally do better.

3. Give the formal definition of Artificial Intelligence.

AI is the ability of a computer to perform tasks commonly associated with intelligent beings, such as reasoning, discovering meaning, generalizing, or learning from experience.

4. Give one real-world example of AI in action.

Self-driving cars use AI to detect obstacles, recognize lanes, and make driving decisions.

5. What human ability do medical image analysis systems try to replicate?

They replicate radiologists’ ability to identify subtle patterns in X-rays or MRI scans.

6. What are perceptual skills in AI?

Perceptual skills involve recognizing and interpreting sensory information such as images, sounds, and patterns.

7. Give two examples of perceptual skills in AI.

- Computer vision (face recognition, OCR)
- Speech recognition (voice assistants)

8. What are linguistic skills in AI?

Linguistic skills involve understanding, generating, and processing human language.

9. Give two examples of linguistic AI applications.

- Machine translation (Google Translate)

- Text summarization systems

10. What are commonsense skills in AI?

Commonsense skills involve applying everyday knowledge, reasoning, and judgment to situations—something humans do naturally.

11. Give one example of commonsense reasoning in AI.

Understanding cause and effect, such as knowing that dropping a glass will likely break it.

12. What is a production system in AI?

A production system is a rule-based framework that uses IF-THEN rules to solve problems through reasoning.

13. Name the three main components of a production system.

- Rule Base
- Working Memory
- Inference Engine

14. What is the role of the working memory?

It stores all facts and data related to the current state of the problem and updates dynamically.

15. What does the inference engine do?

It matches rules with facts, selects appropriate rules, and applies them to derive conclusions.

16. What is conflict resolution in a production system?

It is the strategy used to choose one rule when multiple rules are applicable at the same time.

17. What is forward chaining?

Forward chaining starts with known facts and applies rules to reach a conclusion. It is data-driven reasoning.

18. What is backward chaining?

Backward chaining starts with a goal and works backward to find the rules or facts needed to achieve that goal.

19. What is the goal in the 5L–3L water jug problem?

To measure exactly **4 liters** of water in the 5-liter jug.

20. What is modularity in a production system?

Modularity means each rule is independent, allowing easy updates, maintenance, and scalability of the system.

1. Explain the role of perceptual, linguistic, and commonsense skills in Artificial Intelligence. How do these skills help AI systems behave more like humans?

Artificial Intelligence aims to replicate human intelligence, and this requires mastering three major skill categories: perceptual, linguistic, and commonsense skills.

Perceptual skills allow AI systems to interpret sensory information such as images, sounds, and patterns. Examples include computer vision (face recognition, OCR) and speech recognition (voice assistants). These skills help AI “see” and “hear” like humans.

Linguistic skills involve understanding and generating human language. AI systems use Natural Language Processing (NLP) to interpret grammar, meaning, and context. Applications include chatbots, translation systems, text summarizers, and sentiment analysis tools.

Commonsense skills enable AI to apply everyday reasoning, such as understanding cause-and-effect, social norms, physical constraints, and temporal order. Humans naturally know that fire burns or that breakfast comes before lunch, but AI must be explicitly trained for such reasoning.

Together, these skills help AI systems interact naturally, make decisions, and solve problems in ways that resemble human intelligence.

2. Describe in detail how self-driving cars use Artificial Intelligence. Explain the sensors, decision-making process, and human benchmark comparison.

Self-driving cars are a major application of AI, designed to navigate roads without human input. They use multiple sensors such as **cameras, radar, GPS, and LiDAR** to perceive their surroundings. These sensors collect data about lanes, traffic signs, pedestrians, and nearby vehicles.

AI algorithms process this data using computer vision and pattern recognition to detect obstacles, classify objects, and predict the behavior of other drivers. The decision-making system then determines when to accelerate, brake, or turn.

Humans rely on experience, intuition, and quick reflexes to drive. AI attempts to replicate these abilities by learning from millions of driving examples. In some cases, AI can react faster or more consistently than humans, especially in detecting patterns or anticipating hazards.

Thus, self-driving cars combine perception, reasoning, and action—core elements of AI.

3. Discuss the components of a production system in AI. Explain how the rule base, working memory, and inference engine interact during problem-solving.

A production system consists of three main components:

1. **Rule Base (Knowledge Base):**
Contains IF-THEN rules representing domain knowledge. Each rule encodes expert reasoning.
2. **Working Memory:**
Stores facts about the current problem state. It updates dynamically as rules are applied.
3. **Inference Engine:**
The “brain” that matches rules with facts, selects appropriate rules, and executes them. It uses conflict resolution strategies when multiple rules apply.

Interaction:

The working memory provides the current facts. The inference engine scans the rule base to find rules whose conditions match these facts. Once a rule is selected, its action modifies the working memory. This cycle continues until the goal is reached.

This interaction allows AI systems to reason step-by-step, similar to human logical thinking.

4. Using the water jug problem (5L–3L), explain how a production system applies rules step-by-step to reach the goal state. Include the role of working memory and inference engine.

The water jug problem aims to measure exactly **4 liters** using a 5-liter jug (A) and a 3-liter jug (B). The production system uses rules such as filling, emptying, and transferring water.

Initial State: (0, 0)

Stepwise Execution:

1. Fill Jug A $\rightarrow$ (5, 0)
2. Pour from A to B $\rightarrow$ (2, 3)
3. Empty Jug B $\rightarrow$ (2, 0)
4. Pour A into B $\rightarrow$ (0, 2)
5. Fill Jug A $\rightarrow$ (5, 2)
6. Pour from A to B $\rightarrow$ (4, 3)

Goal State: (4, 0)

Role of Working Memory:

Stores the current state (A, B) after each action.

Role of Inference Engine:

Selects which rule to apply based on the current state and goal. It ensures the sequence of actions leads to the desired outcome.

This example demonstrates how production systems solve problems through rule-based reasoning.

5. What are conflict resolution strategies in production systems? Explain different strategies such as specificity, recency, and priority with suitable examples.

Conflict resolution strategies determine which rule to apply when multiple rules match the current facts.

1. Specificity:

More specific rules are preferred over general ones.

Example:

- Rule A: If fever → flu
- Rule B: If fever AND rash → measles
Rule B is more specific and is chosen.

2. Recency:

Rules that match the most recently added facts are preferred.

Example:

If a new symptom is added to working memory, rules related to that symptom are prioritized.

3. Priority Levels:

Rules are assigned numerical priorities. Higher-priority rules execute first.

Example:

In a medical system, life-threatening conditions may have higher priority.

These strategies ensure logical, consistent, and efficient decision-making.

6. Differentiate between forward chaining and backward chaining. Explain their working principles and give real-world examples where each is used.

Forward Chaining (Data-Driven):

Starts with known facts and applies rules to reach a conclusion.

- Used in medical diagnosis systems.
- Example: Symptoms → Possible disease.

Backward Chaining (Goal-Driven):

Starts with a goal and works backward to find supporting facts.

- Used in troubleshooting systems.
- Example: Device not working → Check possible causes.

Difference:

Forward chaining moves from facts to conclusions, while backward chaining moves from conclusions to facts.

Both methods are essential for expert systems and automated reasoning.

7. Discuss the characteristics of production systems in AI, including modularity, knowledge representation, inference-driven reasoning, and deterministic vs. non-deterministic behavior.

Production systems have several defining characteristics:

Modularity:

Rules are independent units. Adding or modifying one rule does not affect others.

Knowledge Representation:

Knowledge is stored as IF-THEN rules, making it easy to understand and update.

Inference-Driven Reasoning:

The inference engine applies logical reasoning to derive conclusions from facts.

Deterministic Behavior:

Some systems always produce the same output for the same input.

Non-Deterministic Behavior:

Other systems may produce different outputs due to probabilistic reasoning or multiple applicable rules.

These characteristics make production systems flexible, scalable, and suitable for complex decision-making.

8. Explain how production systems are used in real-world applications such as healthcare, finance, industrial automation, and robotics. Provide examples from each domain.

Production systems are widely used across industries:

Healthcare:

Expert systems diagnose diseases by matching symptoms with medical rules.

Finance:

Used for fraud detection, credit scoring, and investment recommendations.

Industrial Automation:

Rules control machine operations, maintenance schedules, and quality checks.

Robotics:

Robots use rules to navigate, avoid obstacles, and perform tasks autonomously.

These applications benefit from the structured, rule-based reasoning of production systems.

9. Describe the step-by-step flow of rule execution in a production system. Explain how facts are stored, rules are matched, conflicts are resolved, and goals are achieved.

The flow of rule execution includes:

1. **Storing Facts:**
Working memory stores initial facts about the problem.
2. **Matching Rules:**
The inference engine scans the rule base to find rules whose conditions match the facts.
3. **Conflict Resolution:**
If multiple rules match, strategies like specificity or priority determine which rule to apply.
4. **Executing the Rule:**
The selected rule updates the working memory.
5. **Repeating the Cycle:**
The system continues matching and applying rules until the goal is reached.

This cycle enables systematic, logical problem-solving.

10. What challenges do AI systems face in acquiring commonsense knowledge? Explain why commonsense reasoning is difficult for machines compared to humans.

Commonsense reasoning is difficult for AI because:

- Humans acquire commonsense through lifelong experience, but AI must be explicitly trained.
- Commonsense involves broad, implicit knowledge about the world, which is hard to encode in rules.
- Many commonsense facts are obvious to humans but not formally documented (e.g., fire burns, heavy objects fall).
- AI struggles with context, ambiguity, and social norms.
- Real-world situations often require flexible reasoning, which rigid rule-based systems cannot easily replicate.

Thus, commonsense remains one of the hardest challenges in AI development.